

# Developmental Stage Classification in Pigs

**A Machine Learning Approach Using Transcriptomic Data**

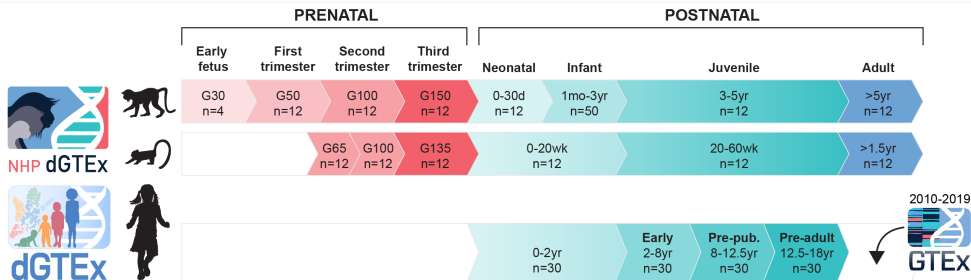
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# 1 Background & Motivation

# The Gap in Human Developmental Data



Comprehensive human developmental transcriptomics remains limited.

Coorens, T.H.H., Guillaumet-Adkins, A., Kovner, R. *et al.* The human and non-human primate developmental GTEx projects. *Nature* 637, 557–564 (2025). doi:10.1038/s41586-024-08244-9

# The Challenge

## Current Gaps

- No molecular standard to stage pigs; chronological age is unreliable.
- Tissue signals differ, so cross-tissue staging is inconsistent.
- Pig-human alignment is unclear; shared biomarkers are underused.

## Our Goal

- **Gene Expression** → **Pig Age**: Define molecular developmental stages from transcriptomics.
- **Pig Age** → **Human Age**: Align pig stages to human timeline via conserved biomarkers.
- **Result**: A unified, cross-species molecular clock.

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## **Data Overview**

# The PigGTEx Project

## Context & Mission

- Part of the **FarmGTEx** consortium (Farm Animal Genotype-Tissue Expression).
- **Goal:** Build a comprehensive catalogue of genetic regulatory variants in pigs.
- Analyze **transcriptomic landscape** across diverse tissues and biological contexts.

## Key Objectives

- **Genetic Mechanisms:** Link genetic variation to gene expression.
- **Biomedical Model:** Leverage pigs as models for human physiology and disease.
- **Sustainable Agriculture:** Inform breeding for health and efficiency.

**A massive public resource:** Integrating thousands of RNA-seq samples to map the pig regulatory genome.

# Data Quality Challenges

## Missing Data

- **52%** of samples missing age information
- **52%** of samples missing sex labels
- **13%** of samples with unknown tissue category

## Age Format Inconsistencies

Age stored as text in various formats:

- “0 day”, “1 day”, “7 days”
- “1 week”, “2 weeks”, “4 weeks”
- “1 month”, “3 months”, “6 months”
- “1 year”, “2 years”
- “unknown”

### Challenge:

Harmonize to a unified numeric format

# Data Processing Pipeline

## 1. Age Harmonization

- Parse text → days
- Map to 5 biological stages:

|                 |              |
|-----------------|--------------|
| Infant          | 0–20 days    |
| Early childhood | 21–59 days   |
| Pre-pubertal    | 60–149 days  |
| Post-pubertal   | 150–365 days |
| Adult           | >365 days    |

## 2. Expression Preprocessing

- $\text{Log}_2(\text{TPM} + 1)$  transformation

## 3. Quality Control

- Filter samples with missing stage
- Explicit “Unknown” sex encoding

## 4. Adaptive Classification

**4-class:**  
Muscle, Liver

**2-class:**  
Lung, Blood, Brain

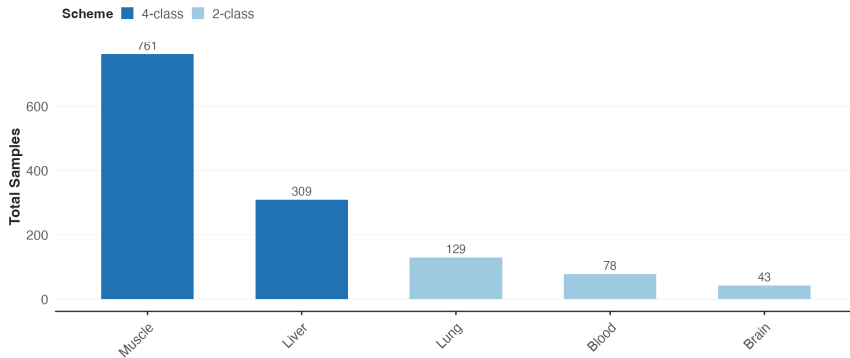
*Tissue-specific schemes ensure sufficient samples per class for ML training.*

↓  
**2,467 curated samples**  
across 8 major tissues

# Classification Schemes

## Classification Schemes by Tissue

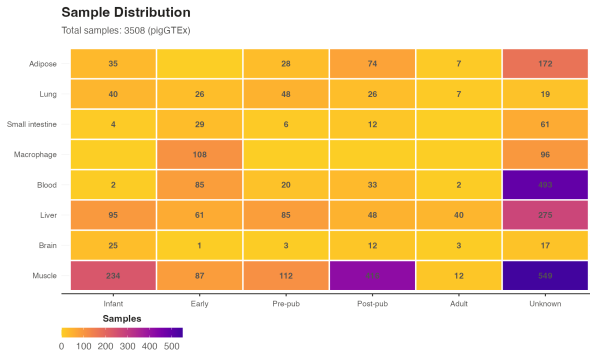
Schemes adapted to sample availability per tissue



# Dataset: pigGTEx

## Scale & Scope

- **2,467 samples** across **8 tissues/cell types**.
- Comprehensive sequencing of:
  - Muscle, Brain, Liver, Blood
  - Intestine, Lung, Adipose, Testis



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## **Methodology**

# Methodology Overview

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## 1. Classification

- Multi-resolution schemes
- 4-class / 3-class / Binary

## 2. Machine Learning

- Ordinal LightGBM
- Binary reduction strategy

## 3. Validation

- Tissue-specific performance
- Cross-tissue robustness

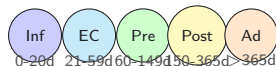
# The Machine Learning Problem

**What is X? (Features)** Gene expression values from RNA-seq

Gene1 = 5.2  
Gene2 = 3.1  
Gene3 = 8.7  
... (20,000+ genes)

Each sample = one pig's gene expression profile

**What is Y? (Label)** Developmental stage we want to predict

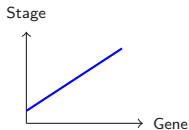


Goal: Gene expression  $\rightarrow$  Predict stage

**In simple terms:** Given a pig's gene activity, which developmental stage is it in?

# Why Machine Learning?

## Traditional Regression



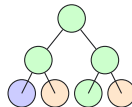
- Assumes linear relationships
- Genes interact non-linearly!

## Deep Learning



- Needs **thousands** of samples
- We only have **hundreds!**

## LightGBM ✓

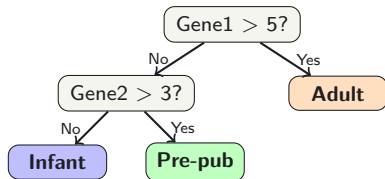


- Handles **non-linear** patterns
- Works with **small samples**

**LightGBM: Best performance with limited sample sizes**

# How Does a Decision Tree Work?

## Simple Example: 3 Genes to Predict Stage



### How it works:

1. Start at the top
2. Ask yes/no questions about genes
3. Follow branches to decision

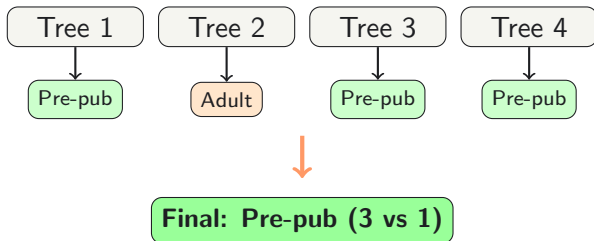
**Example:** Gene1=6.2, Gene2=2.1

- Gene1 > 5? **Yes** → **Adult** (>365d)

A single tree asks simple questions about genes to classify developmental stage.

## From One Tree to Many Trees

One tree can make mistakes. Many trees vote together!



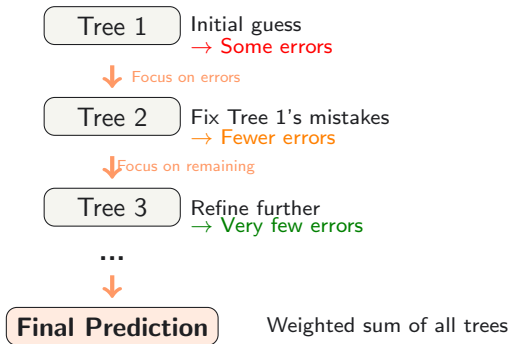
### Why multiple trees?

- Each tree sees data differently
- Mistakes cancel out
- **Majority vote** is more reliable

More trees = More reliable predictions!

# What Makes LightGBM Special: Boosting

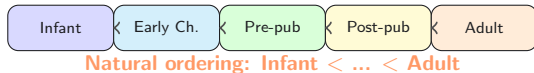
Each new tree learns from previous mistakes



**Boosting:** Trees work as a team, each fixing the others' mistakes

# Why Ordinal Classification Matters

## Developmental Stages Have Order



### Key insight:

Misclassifying Adult as Infant is **worse** than as Post-pubertal!

## Standard vs Ordinal Approach



### Ordinal models:

- Respect stage ordering
- Penalize distant errors more
- Better calibrated probabilities

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## Results

# Performance Metrics Overview

**Precision** Proportion of positive identifications that were actually correct.

$$\text{Precision} = \frac{TP}{TP+FP}$$

**Recall** Proportion of actual positives identified correctly.

$$\text{Recall} = \frac{TP}{TP+FN}$$

**F1 Score** Harmonic mean of Precision & Recall.

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

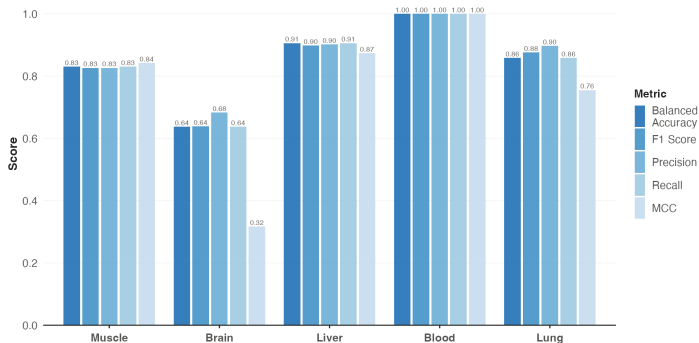
**MCC (Matthews Correlation Coefficient)** Balanced measure for imbalanced datasets.

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$$

# Model Performance

## Model Performance Overview

Multiple classification metrics across tissues



**Strong performance across all metrics: Balanced Accuracy, F1, Precision, Recall, MCC**

# Confusion Matrices

## Confusion Matrices (Selected Tissues)

**Muscle**

|      |     |    |     |      |
|------|-----|----|-----|------|
| Post | 3   | 7  | 0   | 90   |
| Pre  | 3   | 0  | 97  | 0    |
| EC   | 13  | 61 | 4   | 22   |
| Inf  | 85  | 7  | 7   | 0    |
|      | Inf | EC | Pre | Post |

**Brain**

|       |       |      |
|-------|-------|------|
| Late  | 12    | 88   |
| Early | 40    | 60   |
|       | Early | Late |

**Liver**

|      |     |    |     |      |
|------|-----|----|-----|------|
| Post | 0   | 0  | 6   | 94   |
| Pre  | 4   | 0  | 96  | 0    |
| EC   | 0   | 96 | 0   | 4    |
| Inf  | 76  | 0  | 8   | 16   |
|      | Inf | EC | Pre | Post |

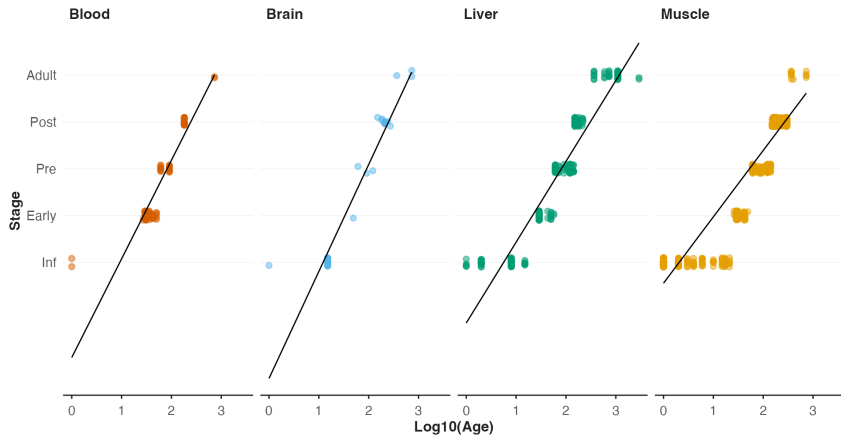
**Blood**

|       |       |      |
|-------|-------|------|
| Late  | 0     | 100  |
| Early | 100   | 0    |
|       | Early | Late |

# Age Correlation

## Correlation: Age vs. Stage Score

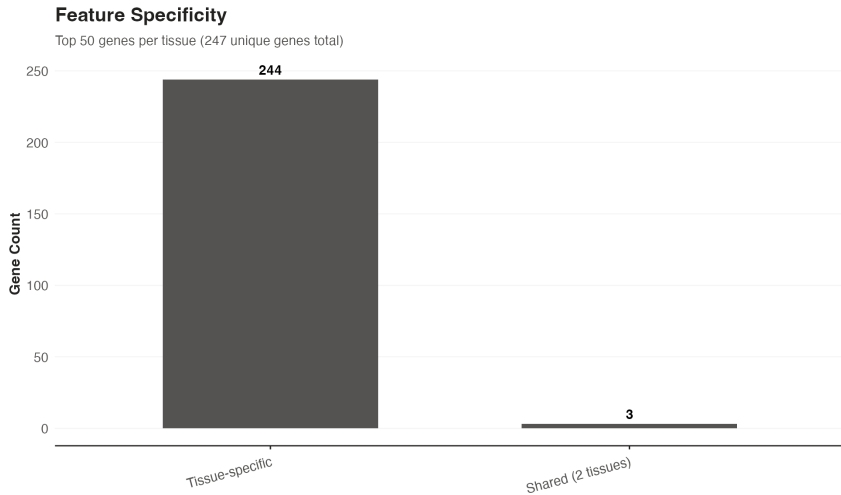
Strong linear relationship between predicted stage and chronological age



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## **Biological Insights**

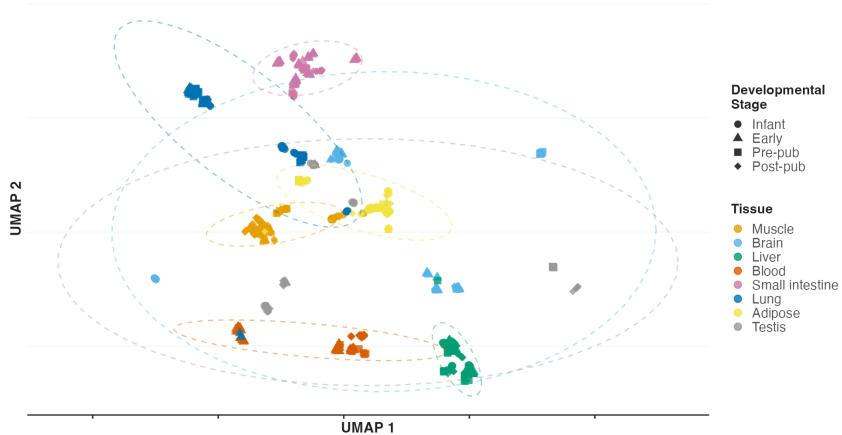
# Feature Overlap Analysis



# Tissue Specificity - UMAP

## Transcriptomic Landscape

UMAP projection of 819 samples showing tissue-specific clustering



# Pathway Enrichment Analysis



## Shared Developmental Programs:

Gene regulation, metabolism, and extracellular matrix organization

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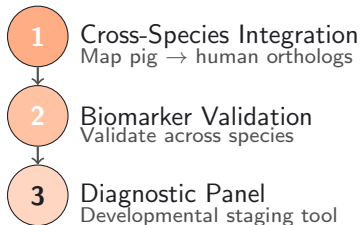
## **Conclusion**

# Summary & Future Directions

## Key Findings

- Accurate staging from transcriptomics
- Tissue-specific molecular clocks
- Conserved developmental programs

## Next Steps



*Lead: Rujing*

**Thank You**